Merge Sort and Divide-and-Conquer: Exercises

All questions are original. Record 3a means key 3 and identity a. Unless specified otherwise, use the chapter's 1-based indices, short-circuit semantics, and word-RAM model.

02-E05 Complete stable MERGE

Merge L=[0a,3b,7c] and R=[1d,3e,6f]. List i,j before each comparison, output prefixes, and the exhaustion tail. Count comparisons, buffer copies, and output writes. If ties choose R instead, what output results, and is it a correct sorting?

02-E06 Sentinels and aliases have assumptions

Legal keys include maximum M. Trace a sentinel implementation using M on L=[M],R=[M,M], with exactly m outputs, through the out-of-bounds read. A second implementation uses a view of A's left half instead of a copy when merging A=[2,4,1,3]. Identify the first destructive overwrite and later bad read. Give a repair without changing the key domain.

02-E07 Odd lengths and peak space

For [6a,2b,5c,1d,4e], list every (p,q,r) and merged result. Count cumulative buffer-copy records, output writes, and maximum records buffered in one merge. Distinguish cumulative allocation, peak buffers, and the call stack.

02-E08 Exact recurrence solutions

For $n = 2^h$, solve $U(1) = 3, U(n) = 2U(n/2) + 5n$ and $W(1) = 0, W(n) = 2W(n/2) + n - 1$ exactly by expansion or induction. Evaluate at 8. W uses the worst comparison count at every merge. Why is it not the exact count for every input?

02-E09 Θ bounds beyond powers of two

Without the master theorem, prove Θ(n log n) for every $n \geq 2$. Specify interval sizes, upper and lower level costs, stopping depth, and total base-case costs. Rounding n up to a power of two and assuming time monotonicity is insufficient.

Merge Sort and Divide-and-Conquer: Exercises (continued)

02-E10 Model changes and an early-exit variant

For sorted input, give insertion-sort and standard merge-sort orders. After recursively sorting the two halves, a variant skips MERGE if $A[q] \leq A[q+1]$. Prove correctness and derive sorted-input time. If a key comparison requires $\Theta(L)$ character reads, which constant-cost conclusions need revision?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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